Elena Rostova

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EXECUTIVE SUMMARY

Founding Systems Architect & Engineering Director with 10+ years architecting autonomous developer infrastructure, distributed runtime systems, and low-latency LLM orchestration. Previously scaled core execution infrastructure at Modal Labs from zero to 120,000 active developers while driving 99.995% compute SLA. Proven operator in zero-to-one product engineering, enterprise developer tooling, and high-velocity developer platforms.

CORE KEY METRICS

$18.4M	120ms	99.995%	45M+
PLATFORM ARR SCALED	MICRO-VM COLD START	EXECUTION SLA	DAILY TOOL INVOCATIONS

CORE COMPETENCIES

Architecture: Autonomous Agent Runtimes, Distributed Systems, Event Streaming, WebAssembly

Languages & Stack: Rust, Go, TypeScript, Next.js, React, Tailwind CSS, Kubernetes, eBPF, gRPC

Technical Leadership: Zero-to-One Product Velocity, Technical Strategy, Developer Platforms, High-Stakes SLAs

PROFESSIONAL EXPERIENCE

VP of Infrastructure & Agent Systems | HyperScale AI — 2023 – Present | San Francisco, CA

- Spearheaded the design and deployment of an event-driven agent orchestration runtime processing 45M tool executions daily across 3,500 enterprise customers.
- Reduced cold-start sandbox latency from 1.8s to 120ms utilizing lightweight micro-VM virtualization and memory snapshotting.
- Grew platform ARR from $1.2M to $18.4M in 18 months by architecting enterprise self-hosted cluster options for Fortune 500 fintech and security teams.
- Built and led an elite 24-person engineering team spanning distributed systems, compiler design, and developer relations.

Lead Systems Architect | Modal Compute — 2020 – 2023 | San Francisco, CA

- Architected the core distributed serverless scheduler handling 250,000 concurrent GPU container bursts with zero scheduling thrash.
- Introduced real-time telemetry streaming pipelines, reducing client-side debugging turnaround time by 64%.
- Co-authored open-source task orchestration protocols adopted by 80+ AI developer communities and 12,000 GitHub stars.

Senior Software Engineer, Platform Core | Datadog — 2017 – 2020 | New York, NY

- Engineered high-throughput distributed tracing pipelines ingesting 3.2 billion metrics per minute.
- Authored custom eBPF probes for kernel-level TCP socket monitoring across multi-tenant Kubernetes clusters.

EDUCATION

Carnegie Mellon University | B.S. in Computer Science & Applied Mathematics — Pittsburgh, PA | 2013 – 2017